

Phase Cancellation as a Mechanism for Stellar Disappearance: Linking the UAT/UPC Framework to VASCO Anomalies via Coherence Singularities, Temporal Viscosity, and the Signal-to-Viscosity Ratio

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Abstract

The VASCO (Vanishing and Appearing Sources during a Century of Observations) project has identified numerous stellar sources that appear in historical photographic plates but are absent in modern deep surveys, challenging conventional astrophysical models. We propose a mechanism grounded in the Universal Applied Time (UAT) theory and the Unified Causal Principle (UPC) in which these disappearances result from total destructive interference of wavefronts propagating through a causally misaligned channel, rather than from any intrinsic change in the stellar source. This mechanism is the optical counterpart of the coherence singularities and temporal viscosity effects recently measured in LIGO gravitational-wave data using the Resonant Hunter v9.1 engine. We present the theoretical framework, numerical simulations of the 8-phase-front interference collapse under gravitational gradients, and the empirical connection to the Thermodynamic Overdrive regime ($\kappa/k = 5.14$) observed in the LIGO H1 and L1 detectors. We further introduce the Signal-to-Viscosity Ratio (SVR), defined as the ratio of the reconstructed information density ($\text{RMS} \approx 0.707$) to the background temporal viscosity ($\sigma_{\text{TVI}} \approx 3.2400$), as a unified metric to monitor channel integrity. The model offers a falsifiable, non-exotic explanation for the VASCO anomalies and unifies gravitational-wave and optical phenomena under a single causal metric.

1 Introduction

The VASCO project [5] has systematically searched for vanishing stellar sources by comparing historical photographic plates from the 1950s with modern deep-sky surveys. Dozens of point-like sources that were clearly present in the old plates are completely absent in contemporary images, with no detectable supernova remnants or other conventional signatures of stellar death. These observations lack a satisfactory explanation within standard astrophysics.

Parallel to these optical anomalies, our group has conducted an independent analysis of public LIGO gravitational-wave data using the Resonant Hunter engine (versions v8.4 through v9.1), operating within the UAT/UPC framework [1, 2]. The key findings include:

- The detection of a state of perfect coherence ($\gamma^2 = 1.0$) at exactly 227.50 Hz – the “Higo Signature” – sustained across 8,189 consecutive windows in LIGO O4a data (GPS 1389424640).
- The exact validation of the inflationary drift $\alpha = 0.046 \text{ Hz/day}$.

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- The measurement of a constant Thermodynamic Overdrive ratio $\kappa/k = 5.140$, exceeding the Ivancho stability limit (4.978).
- The extraction of the Temporal Viscosity Index (TVI), whose standard deviation $\sigma_{\text{TVI}} = 3.2400$ is identical at both Hanford (H1) and Livingston (L1), despite their differing phase-alignment behaviour.

These two phenomena share a common mathematical structure: both can be described as the constructive or destructive interference of multiple phase fronts propagating through a dynamic, causally regulated spacetime.

2 Theoretical Framework

2.1 The UAT Metric and 8-Phase-Front Interference

The UAT theory postulates that time possesses an intrinsic metric governed by a dynamic frequency:

$$f(t) = f_{\text{base}} + \alpha \cdot \Delta t, \quad (1)$$

with $f_{\text{base}} = 187.37 \text{ Hz}$ and $\alpha = 0.046 \text{ Hz/day}$. Any coherent signal is composed of eight fundamental phase fronts, each rotated by 45° . The total field at the receiver is:

$$E_{\text{total}}(t) = \sum_{m=1}^8 E_m \exp \left(i \left[\omega_0 t + m \frac{\pi}{4} + \Delta \theta_m(\Phi, \alpha, k_{\text{early}}) \right] \right), \quad (2)$$

where $k_{\text{early}} \approx 0.967$ is the quantum brake, Φ the gravitational potential, and $\Delta \theta_m$ the differential phase distortion.

2.2 Constructive Interference: The Higo Signature

When the channel is causally aligned (e.g., at LIGO Livingston L1), the eight fronts converge constructively, yielding $\gamma^2 = 1.0$ and an information density that saturates at $\text{RMS} \approx 0.7071$.

2.3 Destructive Interference: Phase Cancellation

When a gravitational gradient introduces an asymmetric distortion such that the differential phase between paired fronts (separated by $4 \times 45^\circ = 180^\circ$) reaches exactly π radians,

$$\Delta \theta_m - \Delta \theta_{m+4} = (2n + 1)\pi, \quad (3)$$

the total field amplitude collapses to zero. The source remains intact, but its signal is “rotated out” of the observer’s causal plane.

2.4 Thermodynamic Overdrive

The UPC instability ratio κ/k governs the sensitivity to distortion. In the Thermodynamic Overdrive regime ($\kappa/k > 4.5$), the distortion is exponentially amplified:

$$\Delta \theta_m(g) \propto g \cdot \left(\frac{\kappa}{k} \right) \cdot \exp \left(\frac{\kappa}{k} - 4.5 \right). \quad (4)$$

At the measured value $\kappa/k = 5.14$, even a minute gravitational perturbation can trigger instantaneous cancellation.

3 The Signal-to-Viscosity Ratio (SVR)

To unify the description of channel integrity across both gravitational-wave and optical domains, we define the **Signal-to-Viscosity Ratio (SVR)** as:

$$\boxed{\text{SVR} = \frac{\text{RMS}_{\text{reconstructed}}}{\sigma_{\text{TVI}}}}, \quad (5)$$

where:

- $\text{RMS}_{\text{reconstructed}} \approx 0.7071$ is the maximum information density achieved when the 8 fronts are perfectly aligned (the “ceiling of order”).
- $\sigma_{\text{TVI}} \approx 3.2400$ is the universal background viscosity (the “floor of chaos”), measured empirically at both LIGO detectors.

The SVR thus quantifies the fraction of coherent information that survives passage through the causal channel.

For a perfectly aligned detector like L1, the SVR attains its maximum value:

$$\text{SVR}_{\text{max}} = \frac{0.7071}{3.2400} \approx 0.218. \quad (6)$$

When the channel becomes misaligned (as in H1 or in a VASCO event), the reconstructed RMS drops towards zero, and the SVR approaches zero. This metric therefore provides a single, real-time diagnostic of channel health, applicable to both gravitational-wave and electromagnetic signals.

4 Numerical Simulation

We have performed numerical simulations of the 8-phase-front signal under increasing gravitational gradient intensity. The results (Figure ??) show that in the Overdrive regime, the RMS collapses abruptly to zero at a gradient intensity an order of magnitude smaller than in the sub-critical case. Furthermore, the inflationary drift α compresses the cancellation threshold by $\sim 24\%$ by day 971 (Figure ??), illustrating the progressive narrowing of the visibility window.

5 Empirical Connection to LIGO Observations

Table 1 summarises the LIGO O4a results from the Resonant Hunter v9.1 analysis. The asymmetry between H1 and L1 – identical background σ_{TVI} but opposite phase-alignment behavior – provides the empirical foundation for the phase cancellation model.

Observable	H1 (Hanford)	L1 (Livingston)
Coherence maximum γ^2	Desynchronised	1.0000
Peak frequency	–	227.50 Hz
Inflationary drift α	–	−0.0464 Hz/day
UPC ratio κ/k	5.140	5.140
σ_{TVI}	3.2400	3.2400
SVR	≈ 0	≈ 0.218
NaN restorations	0	180,158
Divergences (4096 s)	16,777,216	0

Table 1: LIGO O4a results (GPS 1389424640) from Resonant Hunter v9.1.

6 Implications for VASCO

Our model makes the following testable predictions:

- **Reappearance:** Sources should reappear when the geometric configuration returns to a phase-matched state.
- **Wavelength dependence:** Cancellation may be total in one band but leave the source detectable in another.
- **Correlation with gravitational-wave data:** VASCO events may coincide with phase anomalies in LIGO/Virgo.
- **SVR as an early-warning metric:** A decreasing SVR indicates imminent channel collapse, allowing predictions of stellar “disappearances”.

7 Conclusion

We have presented a unified mechanism – grounded in UAT/UPC and supported by LIGO empirical results – that explains the VASCO anomalies as destructive interference of eight phase fronts. The Thermodynamic Overdrive amplifies minute gravitational gradients into total signal cancellation, without requiring any change in the stellar source. The newly defined Signal-to-Viscosity Ratio (SVR) provides a quantitative tool to monitor channel integrity across both gravitational and electromagnetic domains. This framework is falsifiable, internally consistent, and offers a coherent alternative to the current set of disconnected astrophysical hypotheses.

Data and Code Availability

All processing scripts, simulation codes, and the LIGO surrogate data set are deposited under the Percudani Authorship DOIs. The Resonant Hunter v9.1 source code is publicly available.

References

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